

Q.23

A 3-input **majority logic gate** has inputs X , Y , and Z . The output F of the gate is logic '1' if *two or more* of the inputs are logic '1'. The output F is logic '0' if two or more of the inputs are logic '0'.

Which one of the following options is a Boolean expression of the output F ?

(A)	$XY + YZ + ZX$
(B)	$X \oplus Y \oplus Z$
(C)	$X + Y + Z$
(D)	XYZ

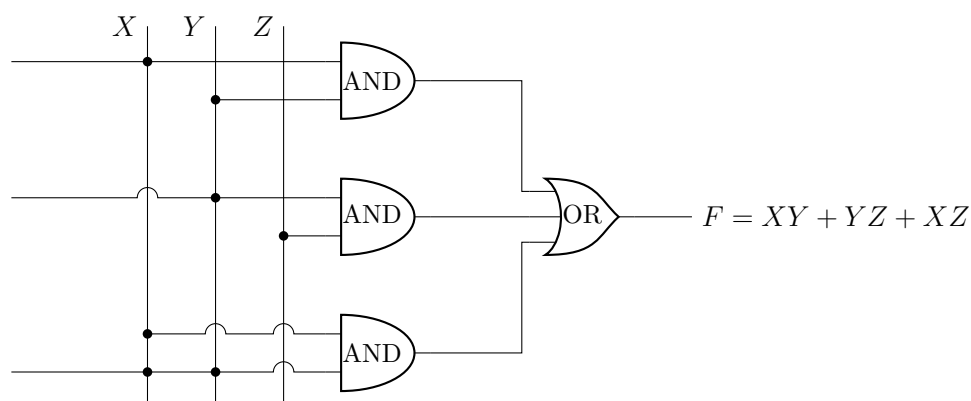
Truth Table for the 3-Input Majority Function

The majority function activates its output whenever at least two of the three inputs are logic 1. The complete truth table is given below.

X	Y	Z	Majority count	F
0	0	0	0 ones	0
0	0	1	1 one	0
0	1	0	1 one	0
0	1	1	2 ones	1
1	0	0	1 one	0
1	0	1	2 ones	1
1	1	0	2 ones	1
1	1	1	3 ones	1

The four minterms where $F = 1$ are: m_3 , m_5 , m_6 , m_7 .

Logic Circuit Diagram of the Majority Gate



Step-by-Step Mathematical Solution

Step 1. Build the Sum of Minterms (Canonical SOP)

From the truth table the function is HIGH for minterms 3, 5, 6, 7:

$$F = \sum m(3, 5, 6, 7) = \bar{X}YZ + X\bar{Y}Z + XY\bar{Z} + XYZ \quad (1)$$

Step 2. Apply Boolean Algebra — Group Pairs

Group XYZ with each of the three other minterms:

$$\text{Group 1: } XY\bar{Z} + XYZ = XY(\bar{Z} + Z) = XY \quad (2)$$

$$\text{Group 2: } X\bar{Y}Z + XYZ = XZ(\bar{Y} + Y) = XZ \quad (3)$$

$$\text{Group 3: } \bar{X}YZ + XYZ = YZ(\bar{X} + X) = YZ \quad (4)$$

Step 3. Combine the Grouped Terms

Using the absorption law $A + A = A$ (the term XYZ is used three times but each pair is valid by the idempotency of Boolean OR):

$$\boxed{F = XY + YZ + ZX} \quad (5)$$

Step 4. Verify Against All Minterms

X	Y	Z	XY	YZ	ZX	$F = XY + YZ + ZX$
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	1	0	1
1	0	0	0	0	0	0
1	0	1	0	0	1	1
1	1	0	1	0	0	1
1	1	1	1	1	1	1

The verification confirms that $F = XY + YZ + ZX$ matches the truth table exactly.

Step 5. Eliminate the Incorrect Options

- **(B) $X \oplus Y \oplus Z$:** XOR of three inputs gives the *odd-parity* function, which outputs 1 for minterms 1, 2, 4, 7 — not the same as the majority function.
- **(C) $X + Y + Z$:** OR of inputs is HIGH as long as *any* input is 1, including the case (0, 0, 1) where the majority is 0. This over-counts.
- **(D) XYZ :** AND of all inputs is HIGH only when *all three* inputs are 1. This under-counts.

Correct Answer

Option (A): $F = XY + YZ + ZX$

This is the standard 3-input **majority function**, implemented with three 2-input AND gates whose outputs feed a 3-input OR gate.

Hardware Implementation Blueprint

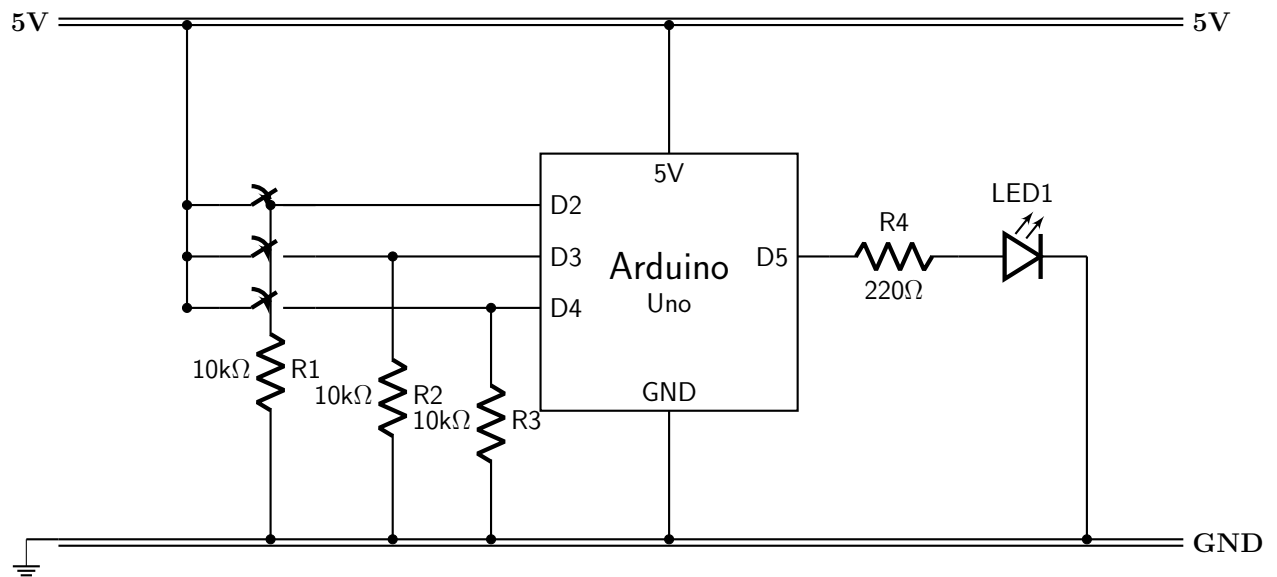
1. Bill of Materials

- 1× Arduino UNO Development Board
- 1× Solderless Breadboard
- 3× Tactile Push Buttons (Inputs X , Y , and Z)
- 3× 10 k Ω Pull-down Resistors
- 1× Light-Emitting Diode (LED — Output F)
- 1× 220 Ω Current-Limiting Resistor
- Solid-core hookup jumper wires
- *Optional:* 74HC08 (Quad 2-input AND) + 74HC32 (Quad 2-input OR) ICs

2. Step-by-Step Pin Mapping & Wiring

- **Input Sub-circuit (Push Buttons):**
 - Bridge Push Button X across the breadboard centre channel. Connect one high-side terminal to the Arduino **5V** rail. Connect the opposite low-side terminal to Arduino **Digital Pin 2**. Tie a 10 k Ω pull-down resistor from Pin 2 to the common **GND** rail.
 - Position Push Button Y identically, wired to **Digital Pin 3**, with its own 10 k Ω pull-down resistor to **GND**.
 - Position Push Button Z identically, wired to **Digital Pin 4**, with its own 10 k Ω pull-down resistor to **GND**.
- **Output Sub-circuit (LED Indicator):**
 - Route a jumper wire from Arduino **Digital Pin 5** to an empty row on the breadboard. Connect the 220 Ω current-limiting resistor from this row to the positive anode leg (longer lead) of the LED.
 - Connect the shorter cathode leg of the LED straight to the common **GND** rail. Complete the loop by linking the breadboard **GND** rail to an Arduino **GND** pin.

3. Hardware Block Diagram



4. Arduino Sketch (Software Logic)

The sketch below reads the three push-button inputs, computes the majority function $F = XY + YZ + ZX$ in firmware, and drives the LED:

Project Source & Links

repo link: <https://github.com/Harshith-Akella-1/gate-q.git>

assembly file link: <https://github.com/Harshith-Akella-1/gate-q/blob/main/gate-q.asm>

arduino files link: <https://github.com/Harshith-Akella-1/gate-q/blob/main/gate-q.ino>

5. Connection Architecture Summary

- **Power Distribution Rails:** The Arduino 5V and GND pins tie directly into the breadboard power rails to deliver stable voltage across all three inputs and the output sub-circuit.
- **Input X Configuration (Pin 2):** Push Button *X* links to Digital Pin 2 with a $10\text{ k}\Omega$ pull-down to GND, ensuring a clean logic 0 when the button is released.
- **Input Y Configuration (Pin 3):** Push Button *Y* hooks into Digital Pin 3 with an identical $10\text{ k}\Omega$ pull-down arrangement.
- **Input Z Configuration (Pin 4):** Push Button *Z* connects to Digital Pin 4 with the same pull-down scheme.
- **Output F Signal Path (Pin 5):** The computed majority result is driven from Digital Pin 5 through a $220\text{ }\Omega$ current-limiting resistor to the LED anode, with the cathode returning to the common GND rail. The LED illuminates whenever $F = 1$, i.e., when two or more of the three inputs are pressed.